

Estimation of Obesity Levels Based on Eating Habits and Physical Condition

Business Application:

Problem Statement: This project aims to estimate obesity levels based on individuals' eating habits and physical conditions.

Industry: Health and Wellness.

Goal: Develop a predictive model that can classify obesity levels, aiding in early identification and intervention.

Objectives: Use machine learning models to predict obesity levels, analyse influential features, and evaluate the models' effectiveness.

Data Characteristics:

Dataset Description: The dataset contains information on eating habits, physical activity, and demographic details related to obesity levels.

Independent Variables: Various attributes capturing lifestyle habits and demographics.

Dependent Variable: Obesity Level (categorized into various classes, such as Insufficient Weight, Normal Weight, Overweight Level I etc.).

Variable Types: Mix of continuous (e.g., physical activity frequency) and categorical variables (e.g., eating habits).

Observations: The dataset includes multiple observations across different obesity levels, with each record representing a unique individual.

Feature Selection:

Most Influential Variables: Feature selection methods like Variance Threshold, RFE (Recursive Feature Elimination), and SelectKBest were used to identify influential variables. These methods helped in narrowing down attributes that significantly impact obesity levels, such as physical activity frequency and dietary habits.

Model Choices and Justifications:

Decision Tree:

Decision Trees are highly interpretable models that allow for clear visualization of the decision-making process. Each node in the tree represents a feature, making it straightforward to understand how each feature contributes to classifying an individual's obesity level. Given that this project deals with a range of obesity classes, a Decision Tree provides a clear and interpretable framework to capture the hierarchical structure in the data.

It's especially useful in scenarios where interpretability is key, as is often the case in health-related predictions. Practitioners can follow the path of decisions (splits) and understand the model's rationale behind classifying a person into a specific obesity level. Additionally, Decision Trees can handle categorical variables effectively without needing extensive preprocessing.

Support Vector Machine (SVM):

SVM is well-suited for classification tasks where the classes are well-separated in a high-dimensional space. SVM works by finding an optimal hyperplane that maximizes the margin between different classes, making it robust in handling complex patterns, especially in multi-class classification tasks like obesity levels.

SVM performs particularly well with clear boundaries between classes, as is the case with extreme obesity categories, where distinct behavioral and physical patterns emerge. The use of RBF kernels (non-linear) also allows SVM to capture non-linear relationships in the data, accommodating complex relationships between dietary habits, physical activity, and obesity levels. This model is a strong choice when accuracy in distinguishing between distinct groups is prioritized, as it minimizes misclassifications for clearly separable classes.

Random Forest:

Random Forest is an ensemble method that builds multiple decision trees and aggregates their results for a more stable and robust classification. By combining trees, it reduces the risk of overfitting, which is often an issue with single decision trees.

In this obesity classification task, there are categories (like "Normal Weight" and "Overweight Level I") that share overlapping feature characteristics. Random Forest's ensemble approach helps smooth out individual tree biases and captures complex interactions between features. It's also useful when certain features are highly influential in specific trees, thus providing a balanced perspective through ensemble voting. Although Random Forests can be less interpretable than single decision trees, their high accuracy and robustness in handling noisy, overlapping data make them a solid choice.

Implementation Overview:

The analysis was implemented in Python using Pandas, NumPy, Matplotlib, and Scikit-learn. The workflow included loading the obesity dataset, handling missing values, encoding categorical variables, preparing the target variable, and splitting the data into training and testing sets for model development and evaluation.

1. Decision Tree with Variance Threshold Feature Selection

Variance Threshold feature selection was applied to remove low-variance features before training the Decision Tree classifier. The model was then evaluated using a classification report, overall accuracy, a confusion matrix, and a tree visualization to understand how feature-based splits contributed to obesity-level classification.

Result:

Decision Tree Classification Report:				
	precision	recall	f1-score	support
Insufficient_Weight	0.86	0.92	0.89	86
Normal_Weight	0.69	0.67	0.68	93
Obesity_Type_I	0.90	0.88	0.89	102
Obesity_Type_II	0.93	0.98	0.96	88
Obesity_Type_III	1.00	0.99	0.99	98
Overweight_Level_I	0.73	0.75	0.74	88
Overweight_Level_II	0.75	0.70	0.72	79
accuracy			0.84	634
macro avg	0.84	0.84	0.84	634
weighted avg	0.84	0.84	0.84	634
Decision Tree Accuracy: 0.8438485804416404				

1. Overall Model Performance

The Decision Tree model does a good job at predicting obesity levels, with an overall accuracy of around **84%**. This means that it correctly classifies about 8 out of every 10 cases, which is pretty strong for a model dealing with different obesity categories.

2. Class-Specific Analysis

Each obesity level has a unique pattern in the data, and the model handles some categories better than others:

Insufficient_Weight:

The model is very good at identifying people who are underweight. With a high **precision (86%)** and **recall (92%)**, it's usually accurate when it predicts someone is underweight and catches most cases in this category.

The **F1-score (0.89)** confirms a strong balance between precision and recall for this class.

Normal_Weight:

The model struggles more with normal-weight predictions, with a lower accuracy (around **69% precision** and **67% recall**). This could be because people with normal weight might share overlapping features with those slightly above or below, making it harder for the model to draw a clear line.

Obesity_Type_I, Obesity_Type_II, and Obesity_Type_III:

For people with higher obesity levels, especially **Obesity Type III**, the model is very confident. It almost never misses cases in these categories, with close to **100% accuracy** for the most severe category. This is likely because people in these categories have distinctive patterns in their data, making them easier to identify.

F1-scores are also high, ranging from 0.89 to 0.99, suggesting that the model has learned to distinguish these categories well, possibly due to distinct features associated with these levels of obesity.

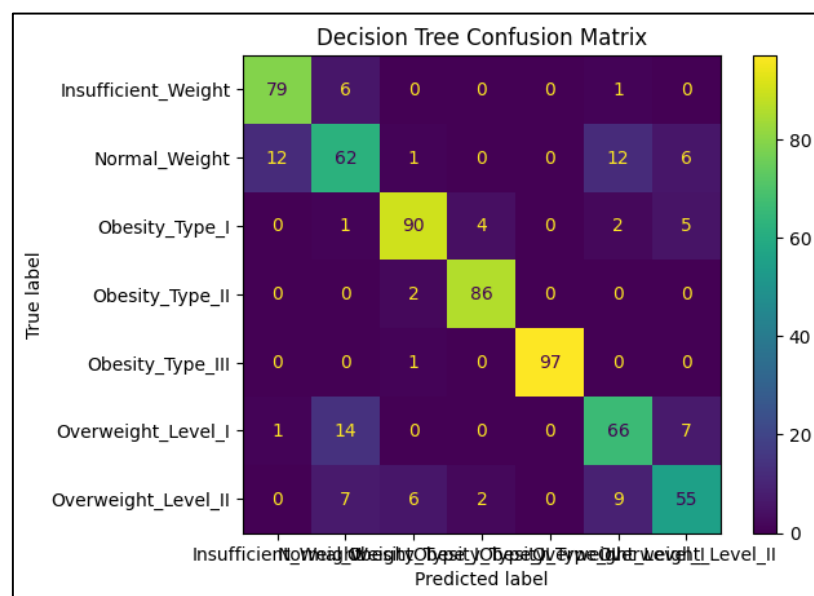
Overweight_Level_I and Overweight_Level_II:

For people in the overweight categories, the model does fairly well, but not as well as it does with obesity types. It's accurate about 70–75% of the time here, which means it can tell when someone is overweight but isn't as certain as it is with more extreme categories.

C. Average Metrics

Macro Average: This metric shows us how well the model performs across all obesity categories equally, without caring about how many people fall into each category. Here, we get an average score of **0.84** for precision, recall, and F1-score. This means that, on average, the model is doing a good job in each category, treating each one as equally important.

Weighted Average: Unlike the macro average, the weighted average takes into account how many people are in each category. So, if a category has more people, it has a bigger influence on the final score. Here, we also get a score of **0.84**, meaning that even when we factor in the differences in category sizes, the model's performance is still strong and consistent across the board.



Confusion Matrix

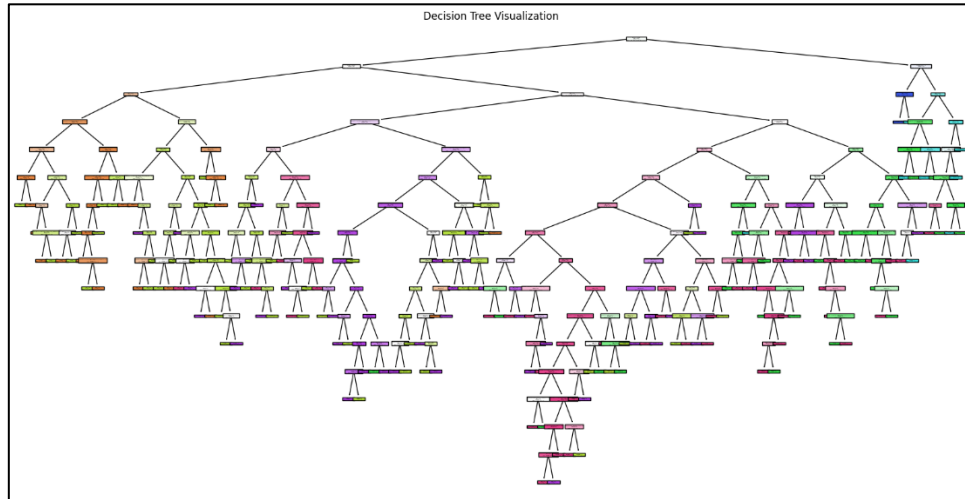
Diagonal cells show correct predictions (e.g., **79** out of **86** for "Insufficient_Weight").

Off-diagonal cells represent misclassifications. For instance:

Normal_Weight is sometimes confused with Insufficient_Weight and Overweight_Level_I.

Overweight_Level_I has several misclassifications as Normal_Weight and Overweight_Level_II.

The model performs well on extreme obesity levels but has more confusion in middle categories, which likely have overlapping features.



Decision Tree Visualization Insights

The tree structure provides insight into the model's decision-making:

Depth and Complexity: The tree is quite deep, with many nodes and branches. This suggests that the model might be overfitting a bit, which means it's picking up on small patterns that may only apply to this dataset and not necessarily to new data.

Feature Splits: Each level of the tree represents a decision based on a feature (like diet habits or physical activity). The model is likely making big distinctions first (e.g., separating extreme obesity from other classes) and then refining further with more detailed splits.

Interpretability: The tree can be interpreted by following paths from the root to leaf nodes. Each path shows a combination of feature-based decisions leading to a specific obesity level classification.

2. Support Vector Machine (SVM) with Recursive Feature Elimination (RFE)

The input features were standardized before applying Recursive Feature Elimination (RFE) to select the two most informative features for the SVM model. A Support Vector Machine with an RBF kernel was trained and evaluated using accuracy, class-level metrics, a confusion matrix, and a decision-boundary plot to examine class separation.

RESULT:

SVM Classification Report:				
	precision	recall	f1-score	support
Insufficient_Weight	0.88	1.00	0.93	86
Normal_Weight	1.00	0.69	0.82	93
Obesity_Type_I	0.98	0.97	0.98	102
Obesity_Type_II	0.96	0.91	0.94	88
Obesity_Type_III	0.93	0.99	0.96	98
Overweight_Level_I	0.80	0.92	0.86	88
Overweight_Level_II	0.90	0.95	0.93	79
accuracy			0.92	634
macro avg	0.92	0.92	0.91	634
weighted avg	0.93	0.92	0.92	634
SVM Accuracy: 0.917981072555205				

Classification Report

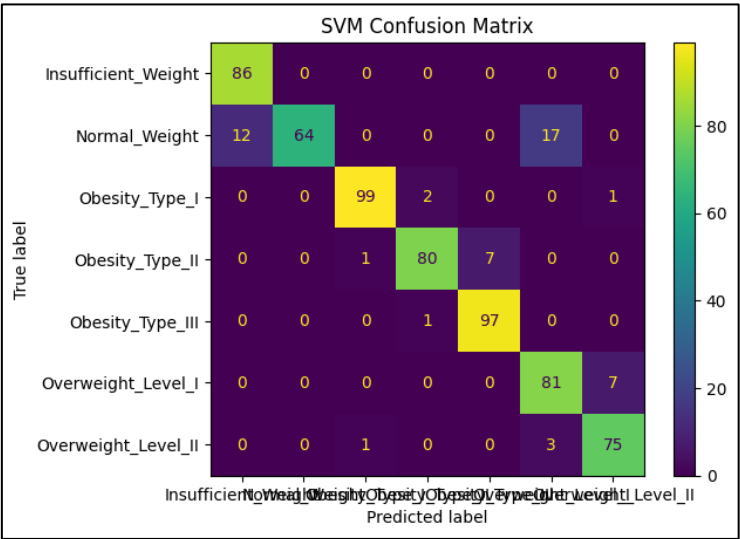
Overall Accuracy: The SVM model achieved an accuracy of 91.8%, meaning it correctly classified nearly 92 out of every 100 cases.

Precision, Recall, and F1-Score:

Precision: Measures the model’s ability to correctly label instances of each category. For instance, "Obesity_Type_I" has a precision of 0.98, meaning nearly all cases labeled as this category were correct.

Recall: Reflects how well the model captures all actual instances of each category. High recall scores (e.g., 1.00 for "Insufficient_Weight") indicate the model successfully identifies these cases.

F1-Score: Balances precision and recall, with scores close to 1 (like 0.96 for "Obesity_Type_III") showing strong performance. The high F1-scores across categories indicate the model is effective overall, with slightly lower performance for "Normal_Weight" (F1-score of 0.82).



Confusion Matrix

Correct Classifications (Diagonal Cells):

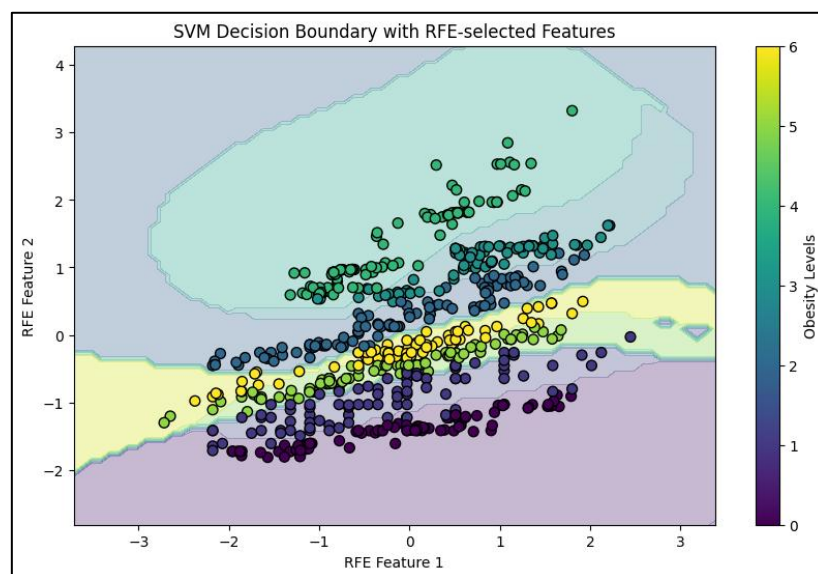
The majority of the predictions are accurate, as seen on the diagonal cells. For example, all 86 cases of "Insufficient Weight" were predicted correctly.

Categories like "Obesity_Type_I," "Obesity_Type_III," and "Overweight_Level_I" show high accuracy with minimal misclassifications.

Misclassifications (Off-Diagonal Cells):

Normal Weight Misclassified: 17 cases of "Normal Weight" were predicted as "Overweight_Level_I," indicating some overlap or similarity in feature values between these categories.

Obesity_Type_II Misclassified: A few cases (7) were incorrectly classified as "Obesity_Type_I," which might suggest slight similarities in the feature distribution for these types.



Decision Boundary Plot:

Distinct Boundaries: The model creates clear boundaries between most classes, indicating it can distinguish different obesity levels effectively based on these features.

Overlap in Middle Categories: There's some overlap in the decision regions for categories like "Normal Weight" and "Overweight Level I," which aligns with the misclassifications seen in the confusion matrix. This overlap could explain why some categories are harder to separate.

3. Random Forest with SelectKBest Feature Selection

SelectKBest was used to identify the top two features for the Random Forest model. The classifier was then trained and evaluated using a classification report, overall accuracy, a

confusion matrix, and a decision-boundary visualization to assess how well the selected features separated the obesity categories.

RESULT:

Random Forest Classification Report:				
	precision	recall	f1-score	support
Insufficient_Weight	0.87	0.90	0.88	86
Normal_Weight	0.71	0.59	0.64	93
Obesity_Type_I	0.71	0.69	0.70	102
Obesity_Type_II	0.80	0.83	0.82	88
Obesity_Type_III	1.00	0.99	0.99	98
Overweight_Level_I	0.55	0.64	0.59	88
Overweight_Level_II	0.51	0.51	0.51	79
accuracy			0.74	634
macro avg	0.73	0.73	0.73	634
weighted avg	0.74	0.74	0.74	634
Random Forest Accuracy: 0.7381703470031545				

Classification Report Analysis

Overall Accuracy: The Random Forest model achieved an accuracy of **73.8%**, meaning it correctly classified around 74 out of every 100 cases.

Performance by Category:

High Accuracy: The model performs exceptionally well for "Obesity_Type_III" with perfect precision (1.00) and very high recall (0.99), indicating that it can accurately identify this category.

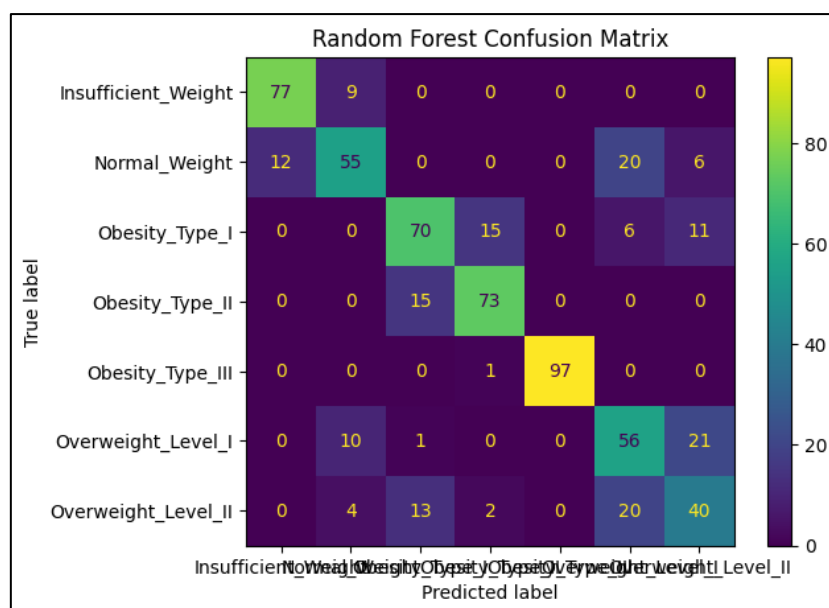
Moderate Accuracy: Categories like "Insufficient Weight" and "Obesity_Type_II" have reasonably good performance, with F1-scores of 0.88 and 0.82, respectively. This suggests the model can usually recognize these categories correctly.

Low Accuracy: The model struggles with "Overweight_Level_I" and "Overweight_Level_II," where F1-scores drop to around 0.51. This indicates that it has a harder time distinguishing these categories, possibly due to overlapping characteristics.

Average Metrics:

Macro Average (0.73): This metric, which averages performance across all categories equally, suggests balanced but moderate accuracy.

Weighted Average (0.74): When taking into account the frequency of each category, the model’s overall performance remains at around **74%**.



Confusion Matrix Analysis

The confusion matrix gives us a more detailed look at where the model is making mistakes.

Correct Classifications (Diagonal Cells):

The model correctly classifies most cases of **"Obesity_Type_III"** (97 out of 98 cases), showing strong performance in this category.

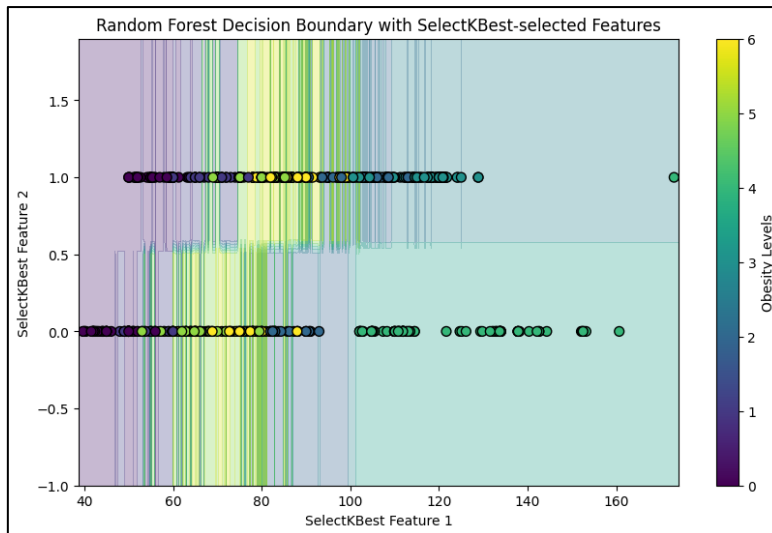
"Insufficient Weight" also shows high accuracy, with 77 out of 86 cases correctly classified.

Misclassifications (Off-Diagonal Cells):

Normal Weight Misclassified: There are **20 cases** of "Normal Weight" predicted as "Overweight_Level_I," indicating that the model struggles to separate these two categories.

Obesity_Type_I and Obesity_Type_II: The model confuses **15 cases** of "Obesity_Type_I" as "Obesity_Type_II," suggesting some overlap in features between these two categories.

Overweight Levels: Both "Overweight_Level_I" and "Overweight_Level_II" have significant misclassifications, showing that the model finds it challenging to distinguish between these closely related classes.



Decision Boundary Plot

The decision boundary plot illustrates how the Random Forest model uses the top 2 features selected by SelectKBest to classify obesity levels:

Clear Boundaries for Distinct Categories:

The model forms distinct regions for certain categories, especially **"Obesity_Type_III"** and **"Insufficient Weight,"** which aligns with their high accuracy in the classification report.

Complex Boundaries for Overlapping Categories:

The plot shows dense boundary lines for categories like "Normal Weight" and "Overweight Levels," suggesting that these classes overlap significantly and are harder for the model to separate clearly.

Conclusion:

This project developed three models, Decision Tree, SVM, and Random Forest—to classify obesity levels from lifestyle data. The Decision Tree model provided interpretability with an accuracy of 84.38%, while SVM achieved 91.8%, performing well for distinct classes but struggling with overlapping ones. Random Forest, at 73.8% accuracy, handled complex patterns effectively, particularly in distinguishing higher obesity levels, though it encountered challenges with closely related categories. Each model offers unique strengths, making them valuable tools in health and wellness applications for early obesity detection.

Among the evaluated models, Support Vector Machine (SVM) achieved the best overall performance with 91.8% accuracy, making it the most suitable model for obesity classification in this dataset.